A
Project
Report On

DRIVER DROWSINESS DETECTION SYSTEM

Submitted in Partial fulfillment of the
requirements for the degree of

Bachelor of Engineering in
Computer Engineering

By

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CERTIFICATE

This is to certify that, the Project titled

DRIVER DROWSINESS DETECTION SYSTEM

Is a bonafide work done by

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Project Report Approval for B.E.

This is to certify that the project entitled ***"*DRIVER DROWSINESS DETECTION SYSTEM*"*** is a Bonafide work done by ***GANESH BIRADAR, NAYAN CHAUDHARI, OMKAR JAGDHANE*** *and* ***SHREEPAD SALVI*** under
the supervision of **Prof.** ***ASHA GAIKAR*****.** This project has been approved for the award of *Bachelor's Degree in Computer Engineering, University of Mumbai.*

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Declaration

I declare that this written submission represents my ideas in my own words and where other's ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Driver fatigue has emerged as one of the leading contributors to road accidents worldwide, especially in long-haul and night-time driving scenarios. In India, where road safety is a growing concern, it is estimated that drowsy driving accounts for 15–20% of all road accidents. Fatigue reduces alertness, slows reaction time, and impairs judgment, often leading to severe or fatal crashes. However, traditional methods of detecting driver fatigue—such as self- reporting or periodic monitoring—are reactive and unreliable.

To address this issue, Driver Drowsiness Detection Systems (DDDS) are being developed to identify early signs of fatigue and alert the driver in real-time. These systems primarily rely on visual indicators such as eye closure rate, blinking frequency, and yawning, which can be detected using computer vision and facial landmark recognition techniques. More advanced implementations may also incorporate physiological data (like heart rate) or vehicle-based metrics (like steering patterns), but image-based systems offer a cost-effective and non-intrusive solution suitable for mass adoption.

This report presents the design and working principles of a vision-based DDDS, which uses Eye Aspect Ratio (EAR) as the key metric for drowsiness detection. By continuously monitoring the driver's eye movements through a webcam and analyzing facial landmarks, the system can detect prolonged eye closure—an early sign of drowsiness. Upon detection, it triggers an alert mechanism (audio or visual) to regain the driver's attention, thereby reducing the risk of accidents.